⭐ LEVEL 4 – 120 QUESTIONS

(Descriptive, reasoning, long-answer style)

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1. Prove that every equivalence relation on a set A partitions A into mutually disjoint equivalence classes.

2. Give an example of a relation that is symmetric and transitive but not reflexive. Justify by verifying each property.

3. Explain why every function is a relation but not every relation is a function.

4. A relation R on integers is defined by:
$aRb \iff a - b$ is even.
Show that R is an equivalence relation.

5. Determine whether the relation $R = \{(a, b) : a \leq b\}$ on real numbers is reflexive, symmetric, or transitive. Give reasoning.

6. Show that the identity relation on any set is both reflexive and symmetric, but not necessarily universal.

7. For the function $f(x) = \sqrt{x^2 + 1} - x$, determine its domain and range by using algebraic manipulation.

8. Let $f : \mathrm{R} \to \mathrm{R}$ be defined as $f(x) = x^3 + 1$. Show that f is bijective, hence invertible.

9. If a function is many–one but onto, explain why it cannot be invertible.

10. Show that if f and g are one-one functions, then $g \circ f$ is also one-one.

11. Is the function $f(x) = |x|$ invertible on $(-\infty, \infty)$? Explain.
Also find an interval on which it *is* invertible.

12. Prove that the composition of two onto functions is onto.

13. For which values of k does the function $f(x) = \sqrt{k - x}$ have real domain? Determine domain explicitly.

14. Show that the range of $f(x) = x + \frac{1}{x}$ for $x > 0$ is $[2, \infty)$.

15. For the relation $R = \{(x, y) : x^2 + y^2 = 25\}$, explain why R is not a function.

16. Determine whether the relation “is parallel to” on the set of all lines in a plane is an equivalence relation.

17. If $f(x) = \frac{x-1}{x+1}$, show that its range excludes exactly one real number. Identify that number.

18. Let $f(x) = \sqrt{1 - x^2}$. Discuss whether it is one-one, onto, or invertible on:

(i) $\mathbb{R}$
(ii) $[-1, 1]$
(iii) $[0, 1]$

19. Prove that the inverse of a bijective function is also bijective.

20. For the piecewise function

$$f(x) = \{\begin{matrix} x+2 & x<0 \\ x^2 & x \geq 0 \end{matrix}$$

Determine domain, range, continuity at x=0, and whether the function is one-one.

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21. Show that the set of all functions from A to B has cardinality $|B|^{|A|}$.

22. If f is non-negative and even, show that f(x)=f(−x). Give an example where the function is not even but still satisfies f(0)=0.

23. Determine the range of $f(x) = \cos(\sin x)$ precisely.

24. Explain why the greatest integer function is not invertible on ℝ.

25. If a relation R on A satisfies $(a, b) \in R \Rightarrow (b, a) \in R$, what can you say about R? Give examples.

26. For the function $f(x) = \frac{1}{\sqrt{x-3}}$, determine:

(i) domain
(ii) range
(iii) whether it is one-one

27. Show that the relation “divides” on natural numbers is reflexive and transitive but not symmetric.

28. Consider the function $f(x) = x^2 - 4x + 6$. Find its minimum and hence derive its range.

29. Prove that the composition of two invertible functions is invertible. Also find the inverse of the composite.

30. For the function $f(x) = \ln(\ln x)$, determine domain and sketch the behavior of f using monotonicity.